

Framing Dembski and Ewert's Specification Model for Design Detection within Montañez's Canonical Specified Complexity Formalism

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Abstract

William Dembski and Winston Ewert developed a formulation of specified complexity grounded in the minimal description length of a pattern or outcome. Their approach can be framed within George Montañez's framework of canonical specified complexity (CSC), which illuminates the underlying logic of their model and of design detection more broadly. This integration also highlights the general applicability of specified complexity as a tool for design detection. Dembski and Ewert's methodology for hypothesis testing is applied to bacterial chemotaxis to illustrate its use in a biological context.

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INTRODUCTION

A central focus of the intelligent design research program has been developing methods to detect in patterns and artifacts evidence of intelligent agency. A primary researcher in this investigation is mathematician and philosopher William Dembski, who devised the first generalized methodology for detecting design, which he detailed in *The Design Inference: Eliminating Chance through Small Probabilities* [1]. He argued that highly improbable patterns that can be referenced by a specification display specified complexity. An example of specified complexity is the carvings on Mount Rushmore, which are highly improbable (i.e., complex) and are specified to match the faces of four presidents. The presence of high levels of specified complexity indicates that an intelligent agent almost certainly generated the pattern.

Dembski later collaborated with computer scientists Robert Marks and Winston Ewert to refine the concept of specification [2,3] and to apply the design-detection schema to biology [4]. In 2023, Dembski and Ewert published a second edition of *The Design Inference*, in which they generalize and quantify specifications based on the description length of a pattern or outcome [5]. This enhancement addressed the challenge of

whether the concept of specification can be rigorously and generally defined and applied [6]. The expanded framework uses the smallest description length to set an upper bound on the probability that the pattern resulted from natural processes and chance. For example, under their minimal-description-length specified complexity (MDL-SC) formalism, the specification of Mount Rushmore could be “four presidents’ faces,” a three-word English-language description.

A description does not have to be in a natural language. Dembski and Ewert (2023) list the criteria for a valid language as “binary, prefix-free, and Turing complete.” In simple terms, a language must be expressible in bits, allow unambiguous descriptions, and be capable of expressing any computable pattern, i.e., one that can be generated by an algorithm.

Another seminal approach to understanding and quantifying specified complexity was developed by computer scientist George Montañez [7]. He formulated what he termed canonical specified complexity (CSC) as a general representation of specified complexity. He demonstrated that other formulations of specified complexity, such as semiotic specified complexity and irreducible complexity, can be represented within the

CSC framework. This article demonstrates that Dembski and Ewert's minimal-description-length formulation can also be mapped onto CSC when a specification is expressed in a natural language. The process of connecting these two approaches clarifies the underlying logic of design detection and highlights the broad applicability of specified complexity for generating design inferences. The MDL-SC hypothesis-testing approach is then applied to bacterial chemotaxis.

CANONICAL SPECIFIED COMPLEXITY

Overview

Montañez demonstrated that a general approach to quantifying specified complexity is based on identifying two functions: a probability distribution and a specification function (Figure 1). The probability distribution, $p(x)$, describes the likelihood that an observation, x , resulted from chance and natural processes, which is designated the chance hypothesis. Common models of a chance (null) hypothesis include the uniform distribution, in which each possible outcome is equally probable (e.g., each possible hand in a fair deal of cards), and the normal (Gaussian) distribution, which describes outcomes clustered symmetrically around a mean.

The second function is a specification function that assigns values to a subset of outcomes, designating their significance to the assigner. I will refer to the assigned values as the outcome's specialness to avoid confusing the specification function with Dembski and Ewert's use of the term specification to refer to the description of a pattern. The specification function, $v(x)$, must be a nonnegative, integrable function that satisfies the normalization constraint $v(X) \leq r$, where X denotes the set of all possible outcomes and r is a positive constant. In addition, the specialness of an outcome must be assigned before it is known whether that outcome is realized. Observing an outcome, assigning it a high specialness, and then arguing the outcome resulted from design based on its retrospectively assigned specialness is invalid.

Montañez defined what he termed the *kardis* $\kappa(x)$ as the constant r multiplied by the ratio of $p(x)$ to $v(x)$

$$\kappa(x) := r \frac{p(x)}{v(x)}. \quad (1)$$

The constant r normalizes $v(x)$ such that the sum or integral of $v(x)/r$ for all possible outcomes is less than or equal to 1. The importance of this scaling becomes clear below. For simplicity, I will assume $v(x)$ is already normalized and set r to 1:

$$\kappa(x) := \frac{p(x)}{v(x)}. \quad (2)$$

Montañez defined the specified complexity measure, $SC(x)$, of an outcome as the negative log base 2 of the kardis:

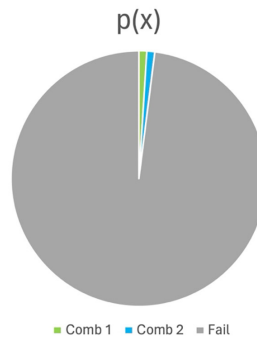
$$SC(x) := -\log_2 \kappa(x), \quad (3)$$

which quantifies specified complexity in bits.

Montañez proved that the probability of the kardis being less than or equal to α is less than or equal to α , assuming the chance hypothesis:

$$P(\kappa(x) \leq \alpha) \leq \alpha. \quad (4)$$

	$p(x)$
Comb 1	1%
Comb 2	1%
Fail	98%



	$v(x)$
Comb 1	50%
Comb 2	50%

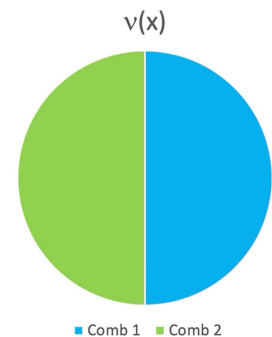


Figure 1. The probability function, $p(x)$, and the specification function, $v(x)$, for a two-dial lock that can be opened with two combinations. Each combination is equally probable if chosen randomly: $p(x) = 0.01$. The two combinations that open the lock are equally special, so $v(x) = 0.5$ for both. The kardis is 0.02, so the chance hypothesis can be rejected for $\alpha > 0.02$. doi:10.5048/BIO-C.2026.1.f1

This relationship can be rewritten in terms of the specified complexity measure:

$$P(SC(x) \geq b) \leq 2^{-b}. \quad (5)$$

The meaning of $\kappa(x)$ is clear in the simple case that only n special outcomes exist, and they have the same $v(x)$, which can be designated v_{sp} :

$$v_{sp} = \frac{1}{n}. \quad (6)$$

The kardis is then the cumulative probability of observing any of the special outcomes. If the probability of observing each of the special outcomes is p_{sp} , then $\kappa(x)$ for any special outcome is $n \cdot p_{sp}$.

In the context of hypothesis testing, the chance hypothesis can be rejected if $\kappa(x) < \alpha$, where α is a probability value that is chosen to provide a desired degree of confidence that the chance hypothesis is false. A lower α results in the chance hypothesis being rejected with greater confidence. The difference from standard hypothesis testing is that the tail does not correspond to extreme values of standard variables such as height or weight, but rather the kardis, which is typically the assigned specialness.

Common choices for α in fields such as psychology and medicine are 0.05 and 0.01. For design detection, it is often chosen to reflect the highest number of events that are possible, meaning that chance is only rejected when the outcome is so unlikely that it could not likely occur even if all known probabilistic resources (e.g., time available, opportunities for trials) were devoted towards producing that outcome.

For example, in the case of a combination lock, the highest number of random trials an individual could realistically attempt is considerably less than 100,000. Based on this estimate, α could be set to 1/100,000 (0.00001) because that value would exhaust the probabilistic resources in realistic contexts.

Similarly, Chatterjee et al. (2014) estimated the highest number of versions of a gene appearing in the history of the Earth to be around 10^{38} [6], so a reasonable choice for α in the context of genetic or protein sequences is 10^{-40} .

Dembski estimated a *universal probability bound* (UPB) based on the age of the universe, the fastest events, and the number of particles in the visible universe. The upper bound on the number of possible events is approximately 10^{150} , yielding a UPB of 10^{-150} [8]. Using such a strategy for setting α can lead to an extremely high degree of confidence for rejecting the chance hypothesis.

Within this formalism, the logic of design detection becomes clear. If $p(x)$ models all physical processes, then an outcome with a sufficiently small kardis can be attributed to intelligent agency with high confidence. Only a mind can choose an outcome so improbable that it could not realistically arise by natural processes and chance, and further, that is designated by a mind as highly special, i.e., assigned a high $v(x)$.

Examples of the Kardis

Examining the meaning of the kardis in the right context further elucidates the underlying logic of Dembski and Ewert's design-detection methodology based on MDL-SC. As mentioned, the kardis is easily interpreted when a set number of outcomes are assigned the same $v(x)$. For example, a four-dial combination lock that can be opened with two combinations (2267 and 1473) has the following probability and specification functions:

$$p(x) = \frac{1}{10^4} \quad (7)$$

$$v(x) = \begin{cases} 0.5 & 2267 \\ 0.5 & 1473 \\ 0 & \text{other} \end{cases} \quad (8)$$

Both combinations are equally special—assigned the same $v(x)$ —since the only concern is opening the lock. The kardis for the combination 1473 directly follows:

$$\kappa(1473) = \frac{p(1473)}{v(1473)} = \frac{0.0001}{0.5} = 0.0002. \quad (9)$$

The kardis equals the probability of randomly picking either combination that opens the lock.

If a set of outcomes has the same $v(x)$ but different $p(x)$, the kardis for a special outcome would be greater than the probability of observing any special outcome having a probability as low or lower than the observed outcome. In the lock example, an observed outcome is a combination that someone attempted to open the lock. If the lock had a defect such that the probability of 1473 being chosen at random is twice as large as other combinations, the probability and kardis would change as follows:

$$p(x) = \begin{cases} \frac{2}{10,001} & 1473 \\ \frac{1}{10,001} & \text{all other combinations} \end{cases} \quad (10)$$

$$\kappa(1473) = \frac{p(1473)}{v(1473)} = \frac{\frac{2}{10,001}}{0.5} \approx 0.0004. \quad (11)$$

In this case, the probability of randomly choosing either combination that opens the lock is approximately 0.0003, which is

comparable to but smaller than the kardis. Similarly, if $p(1473)$ is half the other probabilities, then $p(1473) \approx 0.00005$ and $\kappa(1473) \approx 0.0001$, which is larger. The probability of opening the lock with either combination is approximately 0.00015, and $\kappa(2267) \approx 0.0002$, which is also larger.

If different outcomes with the same $p(x)$ are assigned different $v(x)$, the kardis would be greater than the probability of observing any outcome with the same or higher $v(x)$ than the observed outcome. For example, consider the modified specification function for the combination lock:

$$v(x) = \begin{cases} 0.8 & 2267 \\ 0.2 & 1473 \\ 0 & \text{Other} \end{cases} \quad (12)$$

The combination 2267 might open all locks sold by the manufacturer, so it is assigned a higher $v(x)$. The kardis for combination 1473 increases to 0.0005. It exceeds the probability for randomly opening the lock by a factor of 2.5.

Multiple Categories

If special outcomes fall into N_c categories, the most natural way to define $v(x)$ is to allocate to each category an equal portion, S_c , of the total assigned specialness, S_{tot} , which is the sum or integral of $v(x)$ over all possible outcomes, x :

$$S_{tot} = \begin{cases} \sum_{x \in \mathcal{X}} v(x) & \text{if discrete,} \\ \int_{\mathcal{X}} v(x) & \text{if continuous.} \end{cases} \quad (13)$$

$$S_c = \frac{S_{tot}}{N_c}. \quad (14)$$

As mentioned, $v(x)$ is assumed to be normalized in this analysis, so $S_{tot} = 1$.

Each outcome in each category can then be assigned an equal portion of the specialness assigned to its category. If category C_i has n_i elements, then each outcome in that category has the following specialness:

$$v(x) = \frac{S_c}{n_i} = \frac{1}{n_i N_c} \text{ for } x \in C_i. \quad (15)$$

As an illustration, imagine a nonstandard game of poker where the only winning hands are full houses, four of a kind, and royal flushes. Each hand category is assigned a specialness of $1/3$. The category royal flush includes four hands: royal flush of clubs, diamonds, hearts, and spades. Each of these hands is assigned a specialness of $1/4$ of $1/3$, which equals $1/12$ (Figure 2). The probability that any hand is randomly dealt is 1 chance in the number of combinations of five cards, which is approximately 1 chance in 2.6 million. The kardis for a heart royal flush (HRF) directly follows:

$$\kappa(HRF) = \frac{p(HRF)}{v(HRF)} = \frac{1/\binom{52}{5}}{\frac{1}{12} \cdot \frac{1}{4}} \approx 4.6 \cdot 10^{-6}. \quad (16)$$

If a poker player dealt a heart royal flush as the first hand, one could only conclude cheating with a significance level of approximately 10^{-5} . A significance value of 10^{-6} is smaller than the kardis, so the chance hypothesis could not be rejected with such a high level of confidence. One could not conclude that cheating occurred across all the hands dealt over an entire year in every casino, since that number exceeds 100,000.

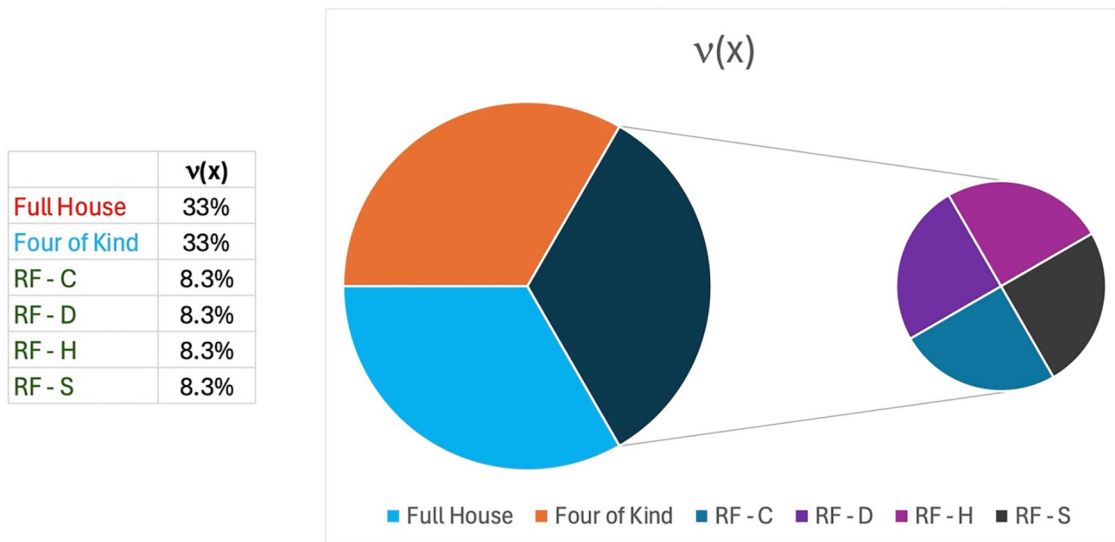


Figure 2. For a game of poker with three special categories (*full house*, *four of a kind*, and *royal flush*), the total specialness is equally divided between them: $v(C_i) = 1/3$, where C_i represents the i th category. The specialness of the royal flush category is equally divided between each of the four specific hands. For the heart royal flush, $v(\text{HRF}) = 1/12$. [doi:10.5048/BIO-C.2026.1.f2](https://doi.org/10.5048/BIO-C.2026.1.f2)

The calculation can be repeated, focusing on the royal flush (RF) category instead of a specific hand within that category:

$$\kappa(RF) = \frac{p(RF)}{v(RF)} = \frac{4/\binom{52}{5}}{\frac{1}{3}} \approx 4.6 \cdot 10^{-6}. \quad (17)$$

As expected, the kardis does not change. The factor of $1/4$ in the specialness changed to a factor of 4 in the probability. The specialness increased by a factor of 4 since the category includes all royal flushes, and the probability of being dealt any royal flush increased by the same factor for the same reason. Focusing on categories is preferred within the MDL-SC framework.

MDL SPECIFIED COMPLEXITY

Overview

Dembski and Ewert generalized specified complexity by framing specifications in terms of minimal description length in a specific language. This approach addresses the question of how to quantify the specialness of specifications if minimal background information is known. Without background knowledge, special categories do not naturally emerge as they do in well-defined contexts, such as card games. In the previous poker example, three special categories of card hands were set (i.e., prespecified) by the rules of the game, which represent the background knowledge. In this context, special hands resulting from design were easy to identify.

The challenge is that many patterns appear in contexts without clear prespecifications. If someone walked into a crowded room, that person would not necessarily know if occupants were playing poker, shooting darts, discussing philosophy, or participating in a fight club. MDL-SC leverages the way languages, by their nature, provide background knowledge and set prespecifications. This feature is easily understood for natural

languages. Cultures assign shorter descriptions to that which they designate as more special.

Outcomes that are highly improbable under the chance hypothesis and have short descriptions are nearly always the product of design. For instance, a poker player dealing four royal flushes in a row is almost certainly cheating. The probability of such an outcome is less than 1 chance in 10^{23} , and the phrase “four royal flushes” is only three words.

In contrast, the description “twenty random cards” is also only three words, but the probability of a set of twenty cards falling into that category is very large. One cannot conclude that such an outcome is designed. Similarly, a specific set of twenty cards represents a highly improbable outcome. Yet, a random set of cards would require a description of numerous words, such as the following: “3 of clubs and spades, 5 of clubs, 7 of diamonds, four jacks, ...” Such a long description could specify many outcomes, so the chance hypothesis cannot be rejected.

Hypothesis Testing with MDL-SC

Dembski and Ewert enumerated the steps for employing MDL-SC in hypothesis testing for design [5]. The steps pertinent to our analysis are the following, given the assumption that specifications are expressed in natural languages:

1. Given a language, an upper bound m on the length of descriptions is chosen that will allow a description to qualify as a specification.
2. The number of specifications (i.e., descriptions no longer than m words), M , is calculated. The number of words used in English and many other natural languages is approximately 100,000, so the number of specifications is approximated as 10^5 multiplied by itself m times:

$$M = (10^5)^m = 10^{5m}. \quad (18)$$

3. The chance hypothesis is rejected with a significance level of α if an event, E , is observed that meets the following conditions: (1) E can be described by a specification V , and (2) the probability that any event with that specification is observed is equal to or less than α/M :

$$P(V') \leq \frac{\alpha}{M}, \quad (19)$$

where V' represents all outcomes with specification V . For the specification “royal flush”, V' represents the four hands in that category: {club 10, J, K, Q, A; diamond 10, J, K, Q, A; heart 10, J, K, Q, A; spade 10, J, K, Q, A}. If the chance hypothesis incorporates all natural processes, rejecting the chance hypothesis equates to a design inference.

As an illustration, the first three hands dealt in a game of cards are all royal flushes. One wishes to determine if the dealer is cheating for $\alpha = 0.01$. One does not know what game is being played, so one cannot identify prespecifications. MDL-SC hypothesis testing proceeds as follows:

1. The number of words in the specification “three royal flushes” is three, so the maximum length of a specification can be set to that number: $m = 3$.
2. The number of possible specifications three words in length immediately follows: $M = (10^5)^3 = 10^{15}$.
3. The probability of being dealt any three royal flushes, $P(V')$, is approximately 3.7×10^{-18} . This number is less than α/M , which is 10^{-17} , so Eq. 19 is satisfied.

The chance hypothesis can be rejected, so the dealer can be accused of cheating since the distribution of card hands should never deviate significantly from the uniform distribution.

What is notable is that the chance hypothesis could not be rejected if only two royal flushes were dealt without having the category *royal flushes* as a prespecification. The probability for two royal flushes is approximately 2.4×10^{-12} , which is larger than α/M . The failure to reject the chance hypothesis in the latter case results from the increased challenge of detecting design without prespecifications, as becomes clear when MDL-SC is connected to CSC.

Connecting to Canonical Specified Complexity

Framing MDL-SC within CSC requires the correct definition of $v(x)$. Within CSC, M represents the number of different categories into which the total specialness is equally divided:

$$v(C_i) = \frac{1}{M} \quad (20)$$

where C_i represents the i th category. The connection between the kardis and Eq. 19 can be derived from the kardis for all the outcomes, V' , referenced by specification V :

$$\kappa(V') = \frac{P(V')}{v(V')} = \frac{P(V')}{1/M} = MP(V') \quad (21)$$

This expression can be plugged into the CSC criterion to reject the chance hypothesis, $\kappa(V') \leq \alpha$, to generate the following inequality: $MP(V') \leq \alpha$, which yields Eq. 19 when both sides are divided by M : $P(V') \leq \alpha/M$.

Detecting design without prespecifications requires envisioning a separate category for every combination of m words. This number is vastly larger than the number of prespecifications in any realistic context—10 special categories of hands in a standard game of poker versus 1,000,000,000,000,000 three-word specifications. The larger number of bins into which the specialness must be divided in MDL-SC results in the probability of observing an event with a given specification, $p(V')$, needing to be much smaller to reject the chance hypothesis. This increase in difficulty is due to a smaller $v(V')$ yielding a larger kardis.

While MDL-SC requires lower probabilities to reject the chance hypothesis, the generalized framework allows rigorous hypothesis testing across diverse contexts. All that is required is to calculate an upper bound on the probability of an outcome under the chance hypothesis and to construct a short specification. Often, identifying short specifications is far easier than unambiguous prespecifications.

The optimal number of words in a specification naturally follows from the number of words in the specification's language, Eq. 18, and Eq. 19. Adding a word to a specification will reduce the kardis if the probability of an outcome falling under the expanded specification reduces by a factor greater than the number of words in the specification's language, W . An additional word reduces $v(V')$ by a factor of W , which is approximately 100,000 in English. For example, adding the word “my” to the specification “financial advisor” reduces the number of people in the corresponding category by a factor greater than 100,000, so the added word leads to a smaller kardis. In contrast, adding the word “heart” to the specification “royal flush” only reduces the number of hands by a factor of four, so adding the word increases the kardis.

Complexity and Specification Measures

The specified complexity measure (Eq. 3) can be separated into measures for the complexity, $C(x)$, and specification, $S(x)$, using the definition of the kardis (Eq. 2):

$$CS(x) := -\log_2 \frac{p(x)}{v(x)} = -\log_2 p(x) + \log_2 v(x) = C(x) - S(x) \quad (23)$$

$$C(x) = -\log_2 p(x), \quad (24)$$

$$S(x) = -\log_2 v(x). \quad (25)$$

The complexity measure is easy to understand if the chance hypothesis entails a set number of outcomes occurring with equal probability. It represents the number of bits required to represent any outcome. If the number of outcomes were 8, the probability for each outcome would be 1/8. The complexity measure of each outcome would then be 3 bits:

$$C(x) = -\log_2 \frac{1}{8} = -\log_2 2^{-3} = 3. \quad (26)$$

Each outcome could be represented by one of the eight three-digit binary numbers: 000, 001, 010, 011, 100, 101, 110, 111. If a million outcomes were possible, the complexity measure for each outcome would be approximately 20 bits since 2^{20} is approximately one million.

The specification measure is also easy to understand if each special outcome were assigned the same $v(x)$. It represents the

number of bits required to represent any special outcome. If four outcomes were special, then $v(x) = 1/4$ for each special outcome. Their specification measure would be 2 bits:

$$S(x) = -\log_2 v(x) = -\log_2 \frac{1}{4} = \log_2 4 = 2. \quad (27)$$

Each special outcome could be represented by one of the four two-digit binary numbers: 00, 01, 10, 11.

The specification measure for special outcomes that fall within multiple categories can be derived from Eq. 15 and Eq. 25:

$$S(x) = -\log_2 \frac{1}{n_i N_c} = \log_2 n_i N_c = \log_2 n_i + \log_2 N_c. \quad (28)$$

As mentioned, n_i is the number of outcomes in the category that contains x , and N_c is the number of categories. The specification measure for x is the number of bits required to represent each outcome in the category that contains x added to the number of bits required to represent each category. In effect, the more special outcomes in a category, and the more special categories, the less special the outcome. Larger n_i and N_c lead to larger $S(x)$, which, when subtracted from the complexity term, result in lower specified complexity.

For MDL-SC, N_c is the number of specifications with a word length no greater than m , which approximates the number of words in the language, W , raised to the power of m . The specification measure immediately follows:

$$S(x) = \log_2 N_c = \log_2 W^m = m \cdot \log_2 W. \quad (29)$$

$S(x)$ is therefore the number of bits represented by a given word in its language times the number of words in the specification. If the language contains approximately 100,000 words, the number of bits required to represent a word is approximately 16, which is close to log base 2 of 100,000. The specification measure approximates 16 times m .

As an example of the use of measures, imagine a coin tossed 100 times where the result alternates between heads and tails. The specification for the pattern is the four words “head-tails repeat fifty” (HT50). The probability, $p(x)$, for any outcome is 1 in 2^{100} . The number of specifications for $m = 4$ is $M = (10^5)^4 = 10^{20}$. The measures for the pattern are the following:

$$C(HT50) = -\log_2 \frac{1}{2^{100}} = 100, \quad (30)$$

$$S(HT50) = -\log_2 \frac{1}{10^{20}} \approx 66, \quad (31)$$

$$SC(HT50) = C(HT50) - S(HT50) \approx 34. \quad (32)$$

The upper bound to the probability that a pattern with such high specified complexity is observed, assuming the chance hypothesis, is given by Eq. 5 for $b = 34$:

$$P(SC(x) \geq 34) < 10^{-10}, \quad (33)$$

which leads for $\alpha > 10^{-10}$ to the rejection of the chance hypothesis.

These measures coincide with the formalism of algorithmic specified complexity (ASC) developed by Dembski, Ewert, and Marks [2,3]. The ASC measure is defined as follows:

$$ASC(X, C, P) = -\log_2 p(X) - K(X|C), \quad (34)$$

where X is an event, C is the event's context or background information, $p(x)$ is the probability for the occurrence of X , and $K(X|C)$ is the Kolmogorov complexity of X given C . The definition is identical to Eq. 23 except $S(x)$ is replaced with $K(X|C)$. Here, the Kolmogorov complexity is the length of the shortest program that generates X when provided with the background information C . Dembski and Ewert (2023) directly relate it to $S(x)$: “the description length is fully formalized and, just as with complexity, characterized in terms of bits.”

Application to Chemotaxis

MDL-SC hypothesis testing can be applied to bacterial chemotaxis. The following represents a preliminary sketch of how the test could proceed.

Bacteria that employ flagellar locomotion require a chemical-detecting navigation system (aka chemotaxis) that can perform several tasks:

1. Sense chemical attractants and/or repellents (A-R).
2. Determine if the concentrations of A-R are increasing, decreasing, or remaining the same.
3. If the concentration of attractants is not increasing or the concentration of repellents is increasing, signal the motor to reverse rotation to change the bacterium's direction of motion.
4. In environments where A-R concentrations vary dramatically, continuously adjust sensor sensitivity to differentiate concentrations over several orders of magnitude.

The minimal components required for chemotaxis have been analyzed based on mathematical modeling [10,11], systems engineering modeling [12], and comparative surveys [13]. All studies suggest that a minimal set of proteins includes the following four or their equivalent:

1. Sensor/receptor (MCP): Detects desirable molecules (attractants) and/or undesirable molecules (repellents).
2. Response Regulator (CheY): When active, binds to the flagellar motor, causing a reversal in rotation and changing the direction of motion.
3. Signaling kinase (CheA): When active, attaches a phosphate to CheY and to CheB (if present) to activate them.
4. Adapter protein (CheW): Couples MCP to CheA by inactivating CheA when attractants bind to, or repellents are released from, MCP.

Bacteria that need chemotaxis to function when A-R concentrations vary over many orders of magnitude also incorporate proteins that continuously adjust MCP sensitivity. Such adaptability is termed robust perfect adaptation (RPA). The following two proteins, or their equivalent, are required to modify sensitivity:

5. CheB: When active, adds a methyl group to MCP to decrease its sensitivity to attractants and/or increase its sensitivity to repellents.

6. CheR: Removes a methyl group from MCP to increase its sensitivity to attractants and/or decrease its sensitivity to repellents.

The MDL-SC hypothesis test does not require demonstrating that every protein in a system is necessary. It is sufficient to identify a small subset of proteins that are essential for minimal function. The test assumes the chance hypothesis that randomizing DNA sequences generate each protein in the subset within a single organism and within the same time window. The test proceeds as follows:

1. The specification for the system can be “nanoscale chemical navigation” or “nanoscale robust perfect chemical navigation,” corresponding to $m = 3$ or $m = 5$. These descriptions are not based on distinctly biological terms. Rather, they represent externally defined specifications that are equally applicable to systems engineering and control theory.
2. The number of specifications for $m = 3$ is approximately $M = (10^5)^3 = 10^{15}$. For $m = 5$, $M = (10^5)^5 = 10^{25}$.
3. The probability that a set of randomized DNA sequences in an individual supports a functional navigation system is less than the probability of n random amino acid sequences corresponding to each of n essential chemotactic proteins: $p(n)$. As mentioned above, a conservative significance level for proteins is $\alpha = 10^{-40}$. Based on these parameters, the chance hypothesis can be rejected for the three-word specification if Eq. 19 is satisfied as follows:

$$p(n) < \frac{\alpha}{M} = \frac{10^{-40}}{10^{15}} = 10^{-55}. \quad (35)$$

It can be rejected for the five-word specification if $p(n) < 10^{-65}$.

An upper bound for $p(n)$ is the product of each protein's proportion of functional sequences (rarity). The rarity of the i -th protein, p_i , is the probability that a random amino-acid sequence will fold into a stable structure that performs the protein's function at a biologically beneficial level. Combining probabilities yields:

$$p(n) = \prod_{i=1}^n p_i = p_1 \cdot p_2 \cdot \dots, \quad (36)$$

$$n = \min \{n' \in \mathbb{N} \mid \prod_{i=1}^{n'} p_i < p_{max}\} \quad (37)$$

where p_{max} is 10^{-55} for three-word specifications and 10^{-65} for five-word specifications.

Satisfying Eq. 35 only requires identifying two proteins where $p_1 \cdot p_2 < 10^{-55}$. Multiple studies suggest that any two proteins that perform tasks as complex as those of proteins 1–4 will meet this criterion [18–21], so the chance hypothesis can be rejected with high confidence.

The same holds true for upgrading basic chemotaxis to achieve perfect robust adaptation. Both CheB and CheR, or their equivalents, perform very complex functions. Each rarity is likely more extreme than 10^{-65} , so their product is almost certainly below this threshold. Again, the chance hypothesis can be confidently rejected, leading to the conclusion that chemotaxis was designed, if other naturalistic explanations can be rejected.

CONCLUSION

Framing MDL-SC within the formalism of CSC yields both an intuitive understanding of the logic behind MDL-SC and design detection in general. It also illustrates the broad applicability of specified complexity to design detection. In addition, applying MDL-SC hypothesis testing to bacterial chemotaxis demonstrates its effectiveness in detecting design in biological systems. It enhances Michael Behe's argument for irreducible complexity entailing design by identifying the minimal number of components required to reject the chance hypothesis of simultaneous appearance, which is often only one or two.

Future studies can rigorously connect rejecting the chance hypothesis with the conclusion of design by ruling out other naturalistic explanations. The common assertion that proteins can be added sequentially can be discredited by demonstrating that at least two proteins must appear simultaneously. This task would include surveying the academic literature on how chemotactic proteins and their genes' regulatory regions must be finely calibrated to operate within tight mathematical constraints. Many studies suggest that proteins 1–4, and often 1–6, must be present and highly specified to meet the exacting criteria for chemotaxis to function at a minimally useful level [15,16].

The argument can be further strengthened by evidence that flagellar systems impose a high energetic and material cost to the cell, reducing survivability when the system is inoperable. Any initial proteins would likely disappear from the population long before the full complement of essential proteins could arise. In fact, some bacteria actively halt production of, degrade, or even eject flagellar proteins when resources are scarce [17]. In addition, many chemotactic proteins must be maintained at high abundance and continuously supplied with energy to support navigation. If chemotaxis or motility were inactive, the loss of these proteins would increase viability.

Another common claim is that proteins in other cellular systems could be co-opted to support navigation. Some proteins share limited similarities with chemotactic proteins [14], so a rigorous analysis must demonstrate that any proposed homologs would require extensive retooling before they could integrate into a nascent navigation system. Moreover, if all other core proteins are assumed to already exist, the proteins under consideration must conform to additional, tight constraints to precisely interface with a specific architecture rather than with any of a wide range of hypothetical navigation systems. A complete analysis would demonstrate how these constraints translate into the origin of chemotactic navigation requiring the emergence from randomly mutating sequences of at least two proteins with rarities that satisfy Eq. 37 for $n = 2$.

Future studies can also apply the MDL-SC methodology to other biological systems, such as DNA replication and protein translation. Dembski and Ewert (2023) recommend applying their formalism to biological systems that are essential and irreducibly complex. Another criterion could be systems that contain design patterns that appear repeatedly in life. For instance, Andrews, Wiley, and Sauro (2024) cataloged 21

design patterns commonly seen in cells [22]. Systems could also be studied that employ the same design patterns seen in human engineering, such as four-bar linkages [23] and control feedback loops [24]. Common design patterns in either biology or human engineering are typically assigned short descriptions, which facilitate easier design detection.

Applying MDL-SC to additional systems based on these criteria will continue to raise the rigor of the argument for design in biology. It will also help quantify the tightness of constraints required for systems to function, helping advance research in fields such as medicine and biomimetics.

- Dembski WA (1998) *The Design Inference: Eliminating Chance through Small Probabilities*. Cambridge University Press (Cambridge).
- Ewert W, Dembski W, Marks RJ (2015) Algorithmic specified complexity in the game of life. *IEEE Trans Syst Man Cybern Syst* 45(4): 584–594 [doi:10.1109/TSMC.2014.2331917](#)
- Dembski WA, Ewert W, Marks RJ (2015) Measuring meaningful information in images: algorithmic specified complexity. *IET Computer Vision* 9(6): 884 – 894 [doi:10.1049/iet-cvi.2014.0141](#)
- Marks RJ, Dembski WA, Ewert W (2017) *Introduction to Evolutionary Informatics*. WSPC (Singapore).
- Dembski WA, Ewert W (2023) *The Design Inference: Eliminating Chance through Small Probabilities*.
- Hurd G, Shallit J, Perakh M, Shanks N, Karsal G, et al. (2006) *Why Intelligent Design Fails: A Scientific Critique of the New Creationism*. Rutgers University Press (New Brunswick, NJ).
- Montanez GD (2018) A Unified Model of Complex Specified Information. *BIOcomplexity* 2018(4) [doi:10.5048/bio-c.2018.4](#)
- Wolpert DH, Macready WG (1996) No Free Lunch Theorems for Optimization. *EEE Transactions on Evolutionary Computation* 1(1): 67.
- Ewert W, Dembski WA, Marks RJ (2014) Algorithmic Specified Complexity. In: Bartlett J, Halsmer D, Hall M, eds., *Engineering and the Ultimate: An Interdisciplinary Investigation of Order and Design in Nature and Craft* Blyth Institute Press. pp 131–149.
- Hamadeh A, Roberts MAJ, August E, McSharry PE, Maini PK, et al. (2011) Feedback Control Architecture and the Bacterial Chemotaxis Network. *PLoS Comput Biol* 7(5): e1001130 [doi:10.1371/journal.pcbi.1001130](#)
- Edgington MP, Tindall MJ (2018) Mathematical Analysis of the Escherichia coli Chemotaxis Signalling Pathway. *Bull Math Biol* 80(4): 758–787 [doi:10.1007/S11538-018-0400-Z/FIGURES/12](#)
- James Johansen CD (2024) Bacterial chemotaxis control process analysis with SysML. *Systems Engineering* [doi:10.1002/SYS.21752](#)
- Rao C V., Kirby JR, Arkin AP (2004) Design and Diversity in Bacterial Chemotaxis: A Comparative Study in Escherichia coli and Bacillus subtilis. *PLoS Biol* 2(2): e49 [doi:10.1371/JOURNAL.PBIO.0020049](#)
- Pallen MJ, Matzke NJ (2006) From The Origin of Species to the origin of bacterial flagella. *Nat Rev Microbiol* 4(10): 784–790 [doi:10.1038/nrmicro1493](#)
- Bitbol AF, Wingreen NS (2015) Fundamental Constraints on the Abundances of Chemotaxis Proteins. *Biophys J* 108(5): 1293–1305 [doi:10.1016/j.bpj.2015.01.024](#)
- Araujo RP, Liotta LA (2018) The topological requirements for robust perfect adaptation in networks of any size. *Nature Communications* 2018 9:1 9(1): 1–12 [doi:10.1038/s41467-018-04151-6](#)
- Kaplan M, Subramanian P, Ghosal D, Oikonomou CM, Pirbadian S, et al. (2019) In situ imaging of the bacterial flagellar motor disassembly and assembly processes . *EMBO J* 38(14) [doi:10.15252/EMBJ.2018100957/SUPPL_FILE/EMBJ2018100957-SUP-0007-MOVIEEV3.ZIP](#)
- Reidharr-Olson JF, Sauer RT (1990) Functionally acceptable substitutions in two α -helical regions of λ repressor. *Proteins: Structure, Function, and Genetics* 7(4): 306–316 [doi:10.1002/prot.340070403](#)
- Axe DD (2004) Estimating the Prevalence of Protein Sequences Adopting Functional Enzyme Folds. *J Mol Biol* 341(5): 1295–1315 [doi:10.1016/j.jmb.2004.06.058](#)
- Tian P, Best RB (2017) How Many Protein Sequences Fold to a Given Structure? A Coevolutionary Analysis. *Biophys J* 113(8): 1719–1730 [doi:10.1016/j.bpj.2017.08.039](#)
- Miller B (2025) A methodology for calculating the rarity of diverse proteins based on functional specificity and thermodynamic stability. *PLoS One* 20(12): e0339572 [doi:10.1371/journal.pone.0339572](#)
- Andrews SS, Wiley HS, Sauro HM (2024) Design patterns of biological cells. *BioEssays* 46(3): 2300188 [doi:10.1002/BIES.202300188](#)
- Burgess S (2021) A review of linkage mechanisms in animal joints and related bioinspired designs. *Bioinspir Biomim* 16(4): 041001 [doi:10.1088/1748-3190/ABF744](#)
- El-Samad H (2021) Biological feedback control—Respect the loops. *Cell Syst* 12(6): 477–487 [doi:10.1016/j.cels.2021.05.004](#)